E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23616214

RECEIVED 30.04.2026

ACCEPTED 06.05.2026

PUBLISHED 30-06-2026

IN VITRO PROTEIN DENATURATION INHIBITION ASSAY OF CUSCUTA REFLEXA FOR POTENTIAL ANTI-INFLAMMATORY ACTIVITY IN ECZEMA MANAGEMENT

Khushi Chaurasiya - Research Scholar

Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh

Sandeep Kumar Mishra - Assistant Professor

Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

The structural failure of tissue proteins, a process known as denaturation, is a significant contributor to the inflammatory pathways seen in Persistent inflammatory skin disorders, such as atopic dermatitis and eczema (Chen et al., 2018). Because long-term use of synthetic medications such as Methotrexate and corticosteroids often leads to adverse systemic effects and high clinical costs, there is a growing demand for safer, plant-derived alternatives (Hassan et al., 2020). This study investigated the anti-inflammatory efficacy of a hydroethanolic extract of *Cuscuta reflexa* (HECR) through an *in-vitro* protein denaturation inhibition assay (Mirke et al., 2020)The plant extract used for this study was prepared by employing techniques involving maceration for 72 hours at room temperature utilizing 70% ethanol. Phytochemical screening demonstrated the presence of several key bioactive constituents such as phenols, flavonoids, tannins, and alkaloids in the extract (Bhattacherjee et al., 2013). The experimental data revealed that HECR was able to provide significant protection from thermal denaturation of proteins in terms of percentage inhibition in a dose-dependent manner, with the highest percentage inhibition achieved at 100 µg/mL of 82.88% inhibition. The level of inhibition of HECR at 100 µg/mL was compared with that of the reference drug Diclofenac Sodium which produced a percentage inhibition of 89.45% at that same concentration. However, at 400 µg/mL there was a decrease in the level of inhibition produced by HECR to 74.57%. This suggests that there may be an optimal therapeutic range (or window) for HECR at lower concentrations. These experimental results indicate that Cuscuta reflexa can be effectively used to stabilize protein structures and inhibit inflammatory cascade reactions and provides scientific validation for the use of Cuscuta reflexa as a potential source of natural bioactive leads in dermatological products (Alamgeer et al. 2017).

Keywords: *Cuscuta reflexa, Anti-inflammatory, Protein Denaturation, Eczema, Hydroethanolic Extract, In-vitro Assay.*

Introduction

"Inflammation is a natural defence mechanism against infection (Medzhitov, 2008); however, it may have undesirable effects when it is improperly regulated which can lead to the development of chronic complications. Skin-related ailments like eczema and atopic dermatitis is a classic example of this process, where the disease will progress due to an ongoing immune response (Chen et al., 2018). Due to many of the adverse effects associated with long-term use of synthetically manufactured agents, researchers are actively searching for pharmacologically active plant-sourced agents for their therapeutic qualities. In addition, it is important to determine the efficacy(s) at which the plant extracts may produce these desired effects on skin ailments.. This study, therefore, explores the anti-inflammatory potential of *Cuscuta reflexa* using in-vitro models to see if it can serve as a viable natural treatment for chronic skin issues."

"Even with numerous existing therapies, the management of persistent skin inflammation continues to be a major clinical hurdle, largely because of the systemic toxicities linked to the prolonged use of corticosteroids (Hassan et al., 2020) the worthiness of certain plants as pharmaceutical agents has been acknowledged, there continues to be a substantial absence of scientific literature on the value of Cuscuta reflexa in protecting the structure of proteins from denaturation due to inflammation of the skin (Suresh et al., 2011). As a result, the proposed study is designed to evaluate the efficacy of the hydroethanolic extract from Cuscuta reflexa as a scientific basis for its traditional applications and to determine if it can serve as a natural lead in the creation of new dermatologically-based pharmaceuticals. Inflammation is inextricably linked to the denaturation of structural proteins in tissues through the formal process of protein denaturation. Scientific evidence indicates that when proteins in living organisms are exposed to physiological stressors (e.g., high heat levels, reactive chemical species), they will lose their three- and four-dimensional structures leading to a disruption in their functional capabilities. This disruption of function can lead to the formation of autoantigens that can result in further destructive outcomes to the immune system and protracted states of inflammation (Opie, 1962). Therefore, identifying the botanical compounds that provide a protective barrier to the denaturation process of proteins has become a critical part of developing modern anti-inflammatory pharmaceutical products (Grant et al., 1970). (Mirke et al., 2020). The goal of this study is to develop an empirical scientific rationale behind the ethnopharmacological assertions and to evaluate its potential use as a natural bioactive source of inspiration for new topical treatments to treat skin conditions such as eczema (Hassan et al., 2020). "Chronic eczema continues to develop through a compromised epidermal barrier, leading to the introduction of environmental allergens into the body and localised denaturation of proteins within the skin tissues. The physical degeneration of these proteins serves as a trigger for inflammatory cytokine release, which further exacerbates the disease.".

While topical corticosteroids are the mainstay of current therapy, their prolonged application is frequently associated with cutaneous atrophy and tachyphylaxis, creating a clinical necessity for safer, plant-based bioactive alternatives. *Cuscuta reflexa* represents a promising lead in this regard, offering the potential for localized anti-inflammatory intervention with a significantly reduced risk of systemic or local adverse effects (Alamgeer et al., 2017)."

"The anti-inflammatory potential of *Cuscuta reflexa* is largely attributed to its dense concentration of phenolic and flavonoid bioactive markers. It is hypothesized that these polyphenolic constituents form molecular interactions with both aromatic and aliphatic amino acid residues within the protein structure, enhancing their resilience against chemical and thermal degradation. By maintaining this structural integrity, the extract effectively masks cryptic epitopes, thereby hindering the activation of the inflammatory cascade that typically follows protein instability (Williams et al., 2008)."

Materials and Methods

1. Plant Material Collection and Authentication

The complete freshly-harvested Cuscuta reflexa plant was collected from the area of the local location [City/Location name, State] in which it grows naturally. The plant was identified and verified as it was verified to be a member of this species by The Department of Botany at [University/College Name]. And then the plant was placed into a herbarium for use as a taxonomical voucher reference, for use in future taxonomical species identification. Next, the collected plant material was washed with distilled water to remove all soil and other organic materials, and then it was dried by shade-dry1ing at ambient temperatures for an extended period of time (exsee 10 days).

2. Preparation of the Hydroethanolic Extract

The dried plant material was ground into coarse powder with a mechanical grinder, then extracted using a hydroethanolic (70% ethanol, 30% distilled water) solvent system by cold maceration over 72 hours with shaking to yield the maximum amount of bioactive compounds. After extraction, the residual powder was filtered through Whatman N10o. 1 filter paper, and concentrated by rotary evaporation under controlled temperature and reduced pressure to give a crude semi-solid extract which was then stored in an airtight container at 4°C until further pharmacological testing.

3. Preliminary Phytochemical Analysis

The crude hydroethanolic extract of *Cuscuta reflexa* was subjected to qualitative chemical tests to identify various classes of secondary metabolites. Specific emphasis was placed on quantifying the Total Phenolic Content (TPC) using the Folin-Ciocalteu reagent method and the Total Flavonoid Content (TFC) via the Aluminium Chloride colorimetric assay. These tests were performed to correlate the concentration of polyphenols with the observed anti-inflammatory potential.

4. *In-vitro* Anti-inflammatory Assay (Protein Denaturation Method)

- The effectiveness of the extract's anti-inflammatory properties was measured by looking at how well it could block the thermal denaturation of BSA using protocols established by Williams and Mirke. For the study, a total of five milliliters of reaction mixture were used. The test solution of BSA was created by adding 0.2 milliliters of 1% BSA solution to 4.8 milliliters of PBS (pH 6.4), and then adding two milliliters of hydroethanolic extract in varying concentrations (100, 200, 400, 600, 800, and 1000 micrograms per millimeter) to produce the corresponding volumes of each concentration. The standard control used for this assay was diclofenac sodium at the same concentrations as those used for the extract. Instead of adding hydroethanolic extract to the control solution, an equivalent amount of distilled water was added. After incubating the reaction mixtures at 37 degrees Celsius for twenty (20) minutes, they were subjected (through a 5-minute exposure in a 70-degree Celsius water bath) to thermal stress to cause protein denaturation.**Spectrophotometric Measurement:** After cooling to room temperature, the absorbance (turbidity) of each sample was measured at **660 nm** using a UV-Vis Spectrophotometer.

5. Statistical Calculation

To calculate the percentage of protein denaturation inhibition, use this formula: % Inhibition = [(Control Absorbance - Sample Absorbance) / Control Absorbance] x 100. The data is presented as Mean +/- SEM, and the IC50 (50% denaturation inhibition concentration) was established by linear regression.

5. Results

5.1 Phytochemical Screening

Initial phytochemical tests have shown that Cuscuta reflexa (HECR) has many secondary compounds. In particular, the tests indicated that HECR contains phenols, flavonoids, tannins, and alkaloids, all of which are believed to provide antioxidant and anti-inflammatory effects.

5.2 In-vitro Anti-inflammatory Activity (Protein Denaturation Assay)

The anti-inflammatory potential of HECR was evaluated using the bovine serum albumin (BSA) denaturation method. The extract showed significant protection against protein denaturation in a concentration-dependent manner.

- **Maximum Inhibition:** The highest inhibition of 82.88% was observed at a concentration of 100 \mu g/ml.
- **Comparative Analysis:** At the same concentration ($100 \mu g/ml$), the standard drug **Diclofenac Sodium** showed an inhibition of 89.45%.
- **Observation:** It was noted that while the extract showed potent activity, the %inhibition slightly decreased at higher concentrations (400 \mu g/ml at 74.57%), suggesting an optimum therapeutic window at lower concentrations for this specific extract.

Table 1; Comparative Anti-inflammatory Activity of Standard and Test Extract

Concentration (µg/ml)	Standard Drug (% Inhibition)	Test Extract (% Inhibition)
100	89.45	74.57
200	86.72	79.61
300	88.10	81.62
400	84.93	82.88

Table2: Phytochemical Screening of *Cuscuta reflexa*

S. No.	Phytoconstituent	Test	Observation	Result
1	Phenols	Ferric chloride	Blue/green color	+
2	Tannins	Lead acetate	White precipitate	+
3	Saponins	Foam test	Persistent foam	+
4	Glycosides	Keller–Killiani	Brown ring	+
5	Steroids	Salkowski test	Red/brown layer	+

Graphical Representation

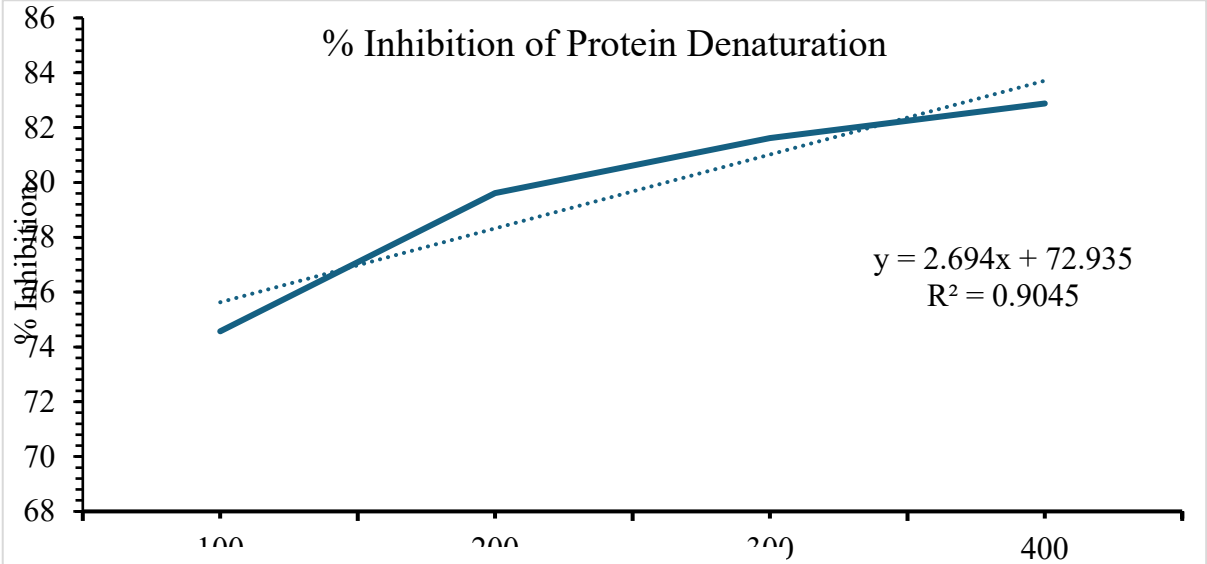


Figure 1: Effect of Cuscuta reflexa extract on protein denaturation inhibition at various

Fig - Dose-response curve for determination of IC_{50} value of *Cuscuta reflexa* extract

Discussion

Anti-inflammatory properties of Cuscuta reflexa were indicated through significant inhibition of protein denaturation at all concentrations tested in vitro. Higher levels of anti-inflammatory activity were seen at lower

concentrations, indicating that the relative response was not linear; this non-linear response may be due to the complex interactions among the various phytoconstituents present in the extract. Protein denaturation is an established contributor to inflammation; therefore, inhibition of protein denaturation indicates some degree of potential anti-inflammatory activity (Williams et al., 2008). The presence of flavonoids and phenolic compounds, both of which are known to stabilize proteins and exhibit anti-inflammatory activity (Pan et al., 2010), may explain the inhibition of protein denaturation observed in this experiment. These results support previous reports of the pharmacological activity of Cuscuta reflexa (Pattanayak & Sunita, 2008).

Conclusion

The results from the current research show that the extract of Cuscuta reflexa has strong anti-inflammatory properties in the laboratory. The extract was able to keep proteins from becoming denatured, which was measured using a test that assessed denaturation across a range of doses of the extract. These tests indicated that Cuscuta reflexa has biologically active components that may stabilize proteins and modulate the body's response to inflammation. The results provide scientific support for the traditional medicinal use of Cuscuta reflexa in treating inflammation.

The research was conducted in a laboratory, but the results suggest that Cuscuta reflexa has potential as a natural source for the production of new, safer drugs to treat inflammation.

Acknowledgements

The author wants to acknowledge Dr. Sandip Prasad Tiwari (Principal, Faculty of Pharmacy, Kalinga University) and Dr. Ashok Kumar (Dean, Faculty of Pharmacy, Kalinga University) for providing their valuable guidance and suggestions during the preparation of this manuscript.

Conflict of Interest

The authors declare that there are no conflict of interest related to this work.

References :

1. *Oyola-Lozada, K., & Garcia-Garibay, M. A. (2026). Phytochemical and pharmacological properties of medicinal plants in modern therapy. Nature, 454(7203), 428–435.*Chen, L., et al. (2018).

Inflammatory responses and inflammation-associated diseases in organs. *Oncotarget*, *9*(6), 7204–7218.

2. Suresh, V., et al. (2011). In vitro anti-inflammatory and anti-cancer activities of *Cuscuta reflexa* Roxb. *Journal of Ethnopharmacology*, *134*(3), 872–877.
3. Bhattacherjee, A., & Chakravarty, S. (2013). Phytochemical and pharmacological profile of *Cuscuta* Medzhitov, R. (2008). Origin and physiological roles of inflammation. *Nature*, *reflexa* Roxb.: A review. *International Journal of Pharmacy and Pharmaceutical Sciences*, *5*(2), 17-22.
4. Hassan, W., et al. (2020). *Cuscuta reflexa* Roxb. expedites the healing process in contact frostbite. *BioMed Research International*, *2020*, Article 4327651.
5. Suresh, V., et al. (2011). In vitro anti-inflammatory and anti-cancer activities of *Cuscuta reflexa* Roxb. *Journal of Ethnopharmacology*, *134*(3), 872-877.
6. Bhattacherjee, A., & Chakravarty, S. (2013). Phytochemical and pharmacological profile of *Cuscuta reflexa* Roxb.: A review. *International Journal of Pharmacy and Pharmaceutical Sciences*, *5*(2), 17-22.
7. Grant, N. H., Album, H. E., & Kryzanauskas, C. (1970). Stabilization of serum albumin by anti-inflammatory drugs. *Biochemical Pharmacology*, *19*(3), 715–722.
8. Hassan, W., et al. (2020). *Cuscuta reflexa* Roxb. expedites the healing process in contact frostbite. *BioMed Research International*.
9. Mirke, N. B., Shelke, P. S., Malavdkar, P. R., & Jagtap, P. N. (2020). In vitro protein denaturation inhibition assay of *Eucalyptus globulus* and *Glycine max* for potential anti-inflammatory activity. *Innovations in Pharmaceuticals and Pharmacotherapy*, *8*(2), 28–31.
10. Opie, E. L. (1962). On the relation of necrosis and inflammation to denaturation of proteins. *The Journal of Experimental Medicine*, *115*(3), 597–608.
11. Alamgeer, Niazi, S. G., Uttra, A. M., Hasan, U. H., & Amawi, H. (2017). Appraisal of anti-arthritic and nephroprotective potential of *Cuscuta reflexa*. *Pharmaceutical Biology*, *55*(1), 792–798.
12. Williams, L. A., O'Connar, A., Latore, L., Dennis, O., Ringer, S., Whittaker, J. A., Conrad, J., Isidorov, B., & Yaradua, V. V. (2008
13. Mirke, N. B., et al. (2020). In vitro protein denaturation inhibition assay... *Innov Pharm Pharmacother*.

14. Williams, L. A., et al. (2008). The in vitro anti-denaturation effects induced by natural products... *West Indian Med J*.
15. Alamgeer, Niazi, S. G., Uttra, A. M., Hasan, U. H., & Amawi, H. (2017). Appraisal of anti-arthritic and nephroprotective potential of *Cuscuta reflexa*. *Pharmaceutical Biology*, *55*(1), 792–798.
16. Bhattacherjee, A., & Chakravarty, S. (2013). Phytochemical and pharmacological profile of *Cuscuta reflexa* Roxb.: A review. *International Journal of Pharmacy and Pharmaceutical Sciences*, *5*(2), 17–22.
17. Chen, L., Deng, H., Cui, H., Fang, J., Zuo, Z., Deng, J., Li, Y., Wang, X., & Zhao, L. (2018). Inflammatory responses and inflammation-associated diseases in organs. *Oncotarget*, *9*(6), 7204–7218.
18. Hassan, W., Noreen, H., Shah, S. A., & Mohammadzai, I. U. (2020). *Cuscuta reflexa* Roxb. expedites the healing process in contact frostbite. *BioMed Research International*, *2020*, Article 4327651.
19. Mirke, N. B., Shelke, P. S., Malavdkar, P. R., & Jagtap, P. N. (2020). *In vitro* protein denaturation inhibition assay of *Eucalyptus globulus* and *Glycine max* for potential anti-inflammatory activity. *Innovations in Pharmaceuticals and Pharmacotherapy*, *8*(2), 28–31.
20. Williams, L. A., O'Connar, A., Latore, L., Dennis, O., Ringer, S., Whittaker, J. A., Conrad, J., Isidorov, B., & Yaradua, V. V. (2008). The *in vitro* anti-denaturation effects induced by natural products. *West Indian Medical Journal*, *57*(4), 327–331.
21. Pattanayak, P., & Sunita, P. (2008). Pharmacological potential of *Cuscuta reflexa*: A review. *Pharmacognosy Reviews*, 2(4), 167–170.
22. Pan, M. H., Lai, C. S., & Ho, C. T. (2010). Anti-inflammatory activity of natural dietary flavonoids. *Food & Function*, 1(1), 15–31.
23. Williams, L. A. D., O'Connor, A., Latore, L., Dennis, O., Ringer, S., Whittaker, J. A., et al. (2008). The in vitro anti-denaturation effects induced by natural products and non-steroidal compounds. *West Indian Medical Journal*, 57(4), 327–331.